

Inside a computer

Under the casing, a computer's hardware is a host of electronic circuitry, components, and connections. As they become ever more powerful, their components need to be smaller, use less power, and generate less heat.

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Components of a computer

The components inside a computer are fairly similar, regardless of the type of computer it is. The parts may look a bit different, but they fulfil the same functions. Understanding what the various parts do and how they work can help users troubleshoot problems or decide whether it is time to upgrade their hardware.

▷ Central processing unit

The central processing unit (CPU), also known as a microprocessor, acts as the brain of the computer. It controls most of the machine's operations and carries out commands. Instructions are sent to the CPU by pressing a key, clicking the mouse, or starting an application, or file.

▷ RAM

This is the system's short-term memory. Whenever a computer performs calculations, it temporarily stores the data in the random-access memory (RAM) until it is needed. The data on the RAM is cleared when the computer is turned off.

The motherboard connects either directly or indirectly to every part of the computer.

The CPU is a ceramic square with a silicon chip located inside.

The RAM is slotted into a reader on the motherboard. RAM can be removed and upgraded if more is needed.

Ports are colour-coded for clear identification.

"...computers have become the most **empowering tool** we've ever created. They're tools of **communication**, they're tools of **creativity**, and they can be **shaped** by their **user**."

Bill Gates (b. 1955), American co-founder of Microsoft

▷ Ports

Computers have an array of ports that allow users to connect external devices to the motherboard. Common ports include universal serial bus (USB), Ethernet (used to connect computers together to form a network), video-graphic array (VGA), high-definition multimedia interface (HDMI), and ports for headphones and microphones.

